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[22BSSS01]

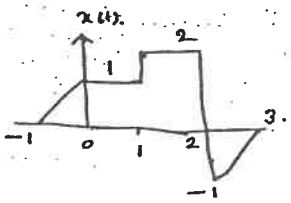
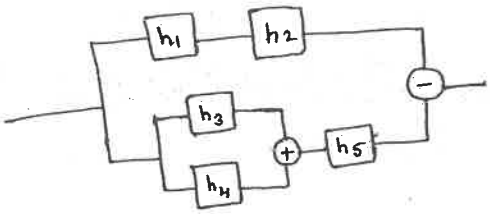
FOURTH SEMESTER B.Tech EXAMINATION, MAY - 2024

SIGNALS AND SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**5 x2=10**
1.	**a.**	Find the odd and even components of the signal $x(n) = [1+n]^2$	L1	1	**02**
	b.	Determine the energy in the following equation $x(t) = e^{3t}$, $0 \le t \le 2$ 0, otherwise	L5	1	**02**
	c.	Evaluate y(t) =x(t) * h(t) x(t) = u(t) h(t) = u(t)	L5	2	**02**
	d.	Find the DTFT of δ(n) and draw the spectrum	L1	5	**02**
	e.	Find Z transform for a given signal $x(n) = a^n u(n)$	L1	6	**02**
		PART – B			**6 X 5=30**
2.	**a.**	Outline x(-3t - 2)  Figure 2a	L2	1	**05**
		OR			
	b.	Determine whether the following system is linear, stable, time invariant, memory, causal. Y(n) = x(2n)	L5	1	**05**
3.	**a.**	Evaluate y(t) = x(t) * h(t) $X(t) = e^{-\lceil t \rceil}$ h(t) = u(t)	L5	2	**05**
		OR			
	b.	Find forced response of given system $5dy(t)/dt + 10\, y(t) = 2x(t)$, $x(t) = e^{-t} u(t)$ Y(0)=0, Y'(0) = 0	L1	2	**05**
4.	**a.**	Find the overall impulse response of the system described by  Figure 4a Where h1= δ(n) + ½ δ(n - 1)	L1	2	**05**

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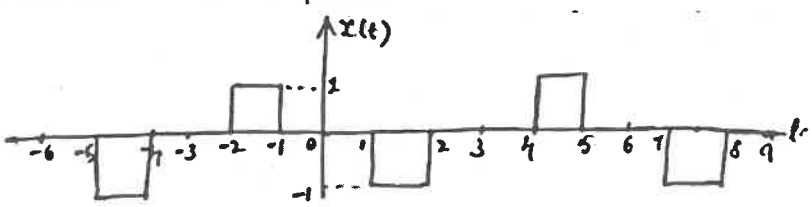
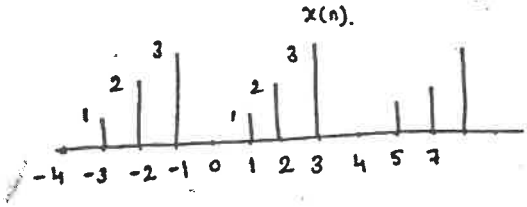
		$h2 = 2\,u(n)$ $h3 = u(n-1) - 2\,u(n)$ $h4 = \delta(n+2)$ $h5 = \delta(n)$			
		OR			
	b.	Compute $y(n) = x(n) * h(n)$ $X(n) = (1/2)^n\,u(n)$ $h(n) = (1/3)^n\,u(n)$	L1	2	**05**
5.	**a.**	State & prove Parseval's theorem property of DTFS.	L5	5	**05**
		OR			
	b.	Evaluate the DTFS representation of signal $x(t) = \sin 4t + \cos 8t$ sketch the magnitude and phase spectrum	L5	4	**05**
6.	**a.**	Determine the Z-transform $x(n) = a^{\lvert n \rvert}$	L5	6	**05**
		OR			
	b.	Define Region of Convergence (ROC) of Z-Transform? List its properties.	L1	6	**05**
7.	**a.**	Find the Fourier transform of the signal $x(t) = e^{-at}\,u(t)$, $a>0$ Draw its magnitude and phase spectrum.	L1	6	**05**
		OR			
	b.	State & prove Parseval's theorem property of DTFS.	L5	4	**05**
		PART – C		**6 X 10=60**	
8.	**a.**	Find the total response for the system given below $y(n) - 3\,y(n-1) - 4\,y(n-2) = x(n) + 2\,x(n-1)$ where $x(n) = 4^n\,u(n)$, $y(-1) = 0$, $y(-2) = 0$	L1	2	**10**
		OR			
	b.	Find the total response for the system given below $d^2y(t)/dt^2 + 5\,dy(t)/dt + 4\,y(t) = dx(t)/dt$ $x(t) = e^{-2t}\,u(t)$, $y(0) = 0$, $y'(0) = 1$	L1	2	**10**
9.	**a.**	Determine the CTFT of the signal $x(t) = e^{-a\lvert t \rvert}$, $a>0$.	L5	5	**10**
		OR			
	b.	Find the DTFT for the following signal $x(n) = a^n\,u(n)$, $\lvert a \rvert < 1$.	L1	5	**10**
10.	**a.**	Evaluate the DTFS representation for the signal $x(n) = \sin(4\pi/21)n + \cos(10\pi/21)n + 1$, sketch the magnitude & phase spectra.	L5	4	**10**
		OR			
	b.	Find the Fourier representation for a periodic impulse train shown in figure below and draw the spectrum.  	L1	4	**10**

		Figure 10b			
11.	**a.**	Evaluate DTFS representation of the signal x(n) shown n figure. sketch the spectrum.  Figure 11a	L5	4	**10**
		OR			
	b.	State & prove the following properties of DTFT. **i.** Convolution property **ii.** Frequency differentiation.	L5	5	**10**
12.	**a.**	Find the inverse Z-transform of x(z). $x(z) = Z / Z^2 - 2Z + 1$ **i.** $\lceil Z \rceil > 1$ **ii.** $\lceil Z \rceil < 1$	L1	6	**10**
		OR			
	b.	Discuss the following properties if Z transform. **i.** Linearity **ii.** Time shift	L6	6	**10**
13.	**a.**	Find the Z-transform of the following **i.** $x(n) = (-2)^n u(n)$ **ii.** $x(n) = (3/4)^n u(n) + 2^n u(-n-1)$	L1	6	**10**
		OR			
	b.	**i.** Determine even and odd components of $x(n) = \{2, 3, -1, 2, 1\}$ **ii.** Determine even and odd components of $x(n) = 1, (-1, 1)\ -2, (1, 2)\ 0$, otherwise	L5	1	**10**